CCD Tiered Evidence and Splink Technical Guide

Status: Management-approved post-POC operating baseline (2026-08-19)
Policy version: `pilot-1.6`
Splink adapter: `pilot-splink-1.1`
Snapshot-specific Review cutoff: `0.938995074`

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1. Executive Summary and Approved Operating Decisions

Management approved the following operating decisions on **2026-08-19** after reviewing POC results and a live demonstration.

1.1 Tiered Gated is the approved deterministic recommendation method

Tiered Gated has **two approved High entry routes**. A pair may satisfy either route, but any trusted-global-identifier conflict still blocks High.

High entry route / condition	Detail
Exact trusted global identifier	A matching trusted global identifier (for example a valid full HKID or HKSR number in global scope) qualifies for High immediately; the result may also carry `name_conflict_warning` if name fields disagree
Exact full Chinese **or** English name **plus** exact independent evidence	Both surname and given name must match exactly after normalization, and at least one independent evidence type (`birthday`, `phone`, or `email`) must also match exactly
No trusted-identifier conflict	Any non-empty trusted global identifier disagreement vetoes High and routes the pair to `Conflict Review`

Passing pairs then proceed through six safety gates in order:

1. **Source-coverage gate** — the source-pair group must be represented in the approved validation cohort
2. **Stale-data gate** — neither record in the pair may have been modified after the frozen snapshot was captured
3. **One-to-many gate** — a record appearing in multiple High pairs within a cluster does not receive automatic recommendation
4. **Transitive-consistency gate** — all High edges in a connected component must form a coherent cluster
5. **Cluster-safety gate** — the full component is checked for internal `Different` labels or conflicts
6. **Data-quality gate** — normalized phone, email, and birthday must pass format validation before becoming evidence

1.2 Passing pairs become reversible recommendations only

Pairs that clear all gates are emitted as **`Proposed`** recommendation records. They do **not**:

- merge CCD Master records;
- set `Is Matched?` in the production Matching Score table;
- modify any existing match flag;
- become active without a separate aggregate-approval step in ERPNext.

The recommendation status lifecycle is:

```
Proposed → (aggregate approval) → Active
         → (reversal)           → Reversed
         → (gate failure)       → Exception
         → (stale-record check) → Stale
```

1.3 Splink is approved only to prioritize optional human review

Splink scores at or above the approved maximum-F1 cutoff are entered into a **ranked human-review queue**. This is an optional, capacity-based pool — not a mandatory backlog.

- A score **at or above** the cutoff means: higher priority for voluntary human review.
- A score **below** the cutoff means: lower review priority. It does **not** mean `Different`.

- No Splink probability threshold qualifies as an automatic High decision.

Current snapshot-specific cutoff: `0.938995074`

This value is specific to the current policy snapshot, `pilot-splink-1.1`, the current 5,000-record training cohort, the current comparison definitions, and the current calibration labels. See Section 26 for when it must change.

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2. Scope and Non-Goals

Included in this guide

- Approved deterministic High path and all six safety gates
- Splink Review queue design, scoring limits, and fail-closed safeguards
- Source mappings, identifier scope, and Reliability Status limitation
- Normalization rules, evidence levels, blocking routes
- Splink Fellegi–Sunter principles, m/u probabilities, and term frequency
- 5,000-record training cohort rationale and constraints
- Calibration vs. training distinction
- Held-out validation methodology
- 20,000-pair batch scoring architecture
- Opaque pair sequences and distinct 5,000 insert chunks
- Snapshot reproducibility checks and fail-closed gates
- Pair-level vs. cluster-level safety
- Human review workflow and role separation
- Approved vs. not-approved decision table
- Sanitized snapshot-specific POC examples
- Recalibration triggers for the snapshot-specific cutoff

Not in this guide

- Record merging or deletion
- Automatic `Is Matched?` updates
- Recall estimates from the High-only validation cohort
- A validated probabilistic High threshold
- Automatic High from partial, masked, or unverified identifiers alone

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3. Governed Sources and Identifier Scope

The pilot governs exactly **10 registered CCD source systems** (referred to as sources S-01 through S-10 in sanitized form). Every source has a versioned `SourceProfile` stored in the policy document.

3.1 SourceProfile structure

```
@dataclass(frozen=True)
class SourceProfile:
    source: str
    field_map: dict[str, str]           # attribute → raw field name
    identifier_scope: dict[str, str]    # attribute → "global" | "local" | "unknown"
    disabled_attributes: frozenset[str] # attributes not applicable to this source
```

3.2 Attribute aliases

The policy resolves canonical attribute names through configurable alias lists:

Canonical attribute	Typical raw field names
`chi_surname`	`chi_surname`
`chi_firstname`	`chi_firstname`
`eng_surname`	`eng_surname`
`eng_firstname`	`eng_firstname`
`phone`	`phone_num`, `res_phone`, `phone`
`email`	`email`, `contact_email`
`birthday`	`birthday`, `dob`
`hksr_num`	`hksr_num`
`hkid`	`hkid`, `hkid_num`

3.3 Identifier scope and trusted global identifiers

An identifier becomes a **trusted global identifier** (a same-person anchor) only when:

1. The policy document lists its canonical attribute in `trusted_global_identifiers`; **and**
2. Both source profiles mark that attribute's scope as `"global"`.

The default scope for any attribute not explicitly configured is `"unknown"`, which is treated as local. A source-specific staff number or client ID uses `"local"` scope and never becomes a cross-source identity anchor.

HKID special rule: A valid HKID requires a full, unmasked value that passes check-digit verification. Partial, masked, or check-digit-invalid HKIDs are downgraded to `unverified_id` scope and cannot be trusted global evidence.

```
Globally comparable  = (attribute ∈ trusted_global_identifiers)
                       AND (left source scope = "global")
                       AND (right source scope = "global")
                       AND (for HKID: valid_hkid(raw_value) = True)
```

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4. Reliability Status Limitation

CCD Master records carry a `reliability_status` field, but the pilot does not use it as a gate on evidence quality. The field was found to be:

- inconsistently populated across source systems;
- not updated when underlying data changes;
- absent from several governed sources.

Using `reliability_status` as evidence or as a gate would silently suppress valid pairs from sources that do not populate the field, and would give false confidence for sources that set it optimistically.

The approved control is the **stale-data gate** (Section 10.2): a snapshot-time comparison detects any record modified after the frozen canary was taken, regardless of the reliability field.

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5. Normalization Rules

All field values pass through normalization before any comparison or blocking-key computation.

Attribute	Normalization applied
Chinese name	CJK normalization, traditional/simplified unification, whitespace stripped, invisible characters removed
English name	Lowercase, accents removed, non-alphabetic stripped, whitespace collapsed
Phone	Digits only, country code stripped if present, leading zeros normalized; **placeholder values removed** (sequential digit strings, all-same digits, non-eight-digit values)
Email	Lowercase, whitespace stripped
Birthday	ISO-8601 `YYYY-MM-DD` after parsing common locale formats
HKID	Uppercase, spaces removed, check digit verified; invalid result is null
Other identifiers	Uppercase, whitespace stripped, dashes removed

Normalization produces `None` (SQL null) for missing or invalid values. A null value means **no evidence**; it is never treated as disagreement. An empty string is treated as null.

The phone normalization specifically removes:
- strings of all identical digits (e.g., `11111111`);
- strings that are fully sequential ascending or descending (e.g., `12345678`);
- strings shorter than 8 digits after cleaning;
- strings containing only non-digit characters.

These are treated as placeholder values that carry no identity information.

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6. Evidence Levels and Cutoffs

Each attribute comparison produces an `Evidence` object with a level from the following ordered enum:

```
class EvidenceLevel(str, Enum):
    MISSING   = "missing"   # one or both sides null after normalization
    DISAGREE  = "disagree"  # both present, do not agree
    WEAK      = "weak"      # some similarity but below thresholds
    CLOSE     = "close"     # substantial similarity, not exact
```

```
    PHONETIC  = "phonetic"  # phonetically equivalent
    EXACT     = "exact"     # string equality after normalization
```

Only `EXACT` satisfies the High-path independent-evidence requirement. `CLOSE`, `PHONETIC`, and `WEAK` contribute to Review routing but never satisfy the High path.

`MISSING` on either side means the field provides no information; it is not counted as `DISAGREE`.

Evidence object properties

```
@dataclass(frozen=True)
class Evidence:
    attribute: str
    level: EvidenceLevel
    score: float              # 0.0–1.0 continuous similarity
    left_present: bool
    right_present: bool
    reason: str               # human-readable comparison explanation

    @property
    def available(self) -> bool:
        return self.left_present and self.right_present

    @property
    def exact(self) -> bool:
        return self.level == EvidenceLevel.EXACT
```

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7. Principle of Tiered Gated

Tiered Gated is the deterministic, evidence-semantic identity decision model used for the approved recommendation path. Unlike Splink, it does **not** learn one compensating probability from all observed patterns at once. It classifies evidence by meaning, applies fixed policy order, and returns a governed operational category.

7.1 What "Tiered" means

Every candidate pair is assigned to one of four operational categories:

Tier	Operational meaning
`High`	Eligible to continue into the six post-decision safety gates for a reversible `Proposed` recommendation
`Review`	Useful evidence exists, but the pair does not satisfy an approved automatic High route
`Conflict Review`	A safety-critical trusted-global-identifier contradiction requires human resolution
`Low`	Evidence is too weak or too sparse to justify automatic action or ordinary review priority

These tiers are **governed actions/categories**, not probability bands. `High` does not mean "high probability" and `Low` does not mean "definitely different"; they are policy outcomes produced by the decision tree.

7.2 What “Gated” means

“Gated” means promotion is constrained by mandatory safety conditions rather than by a freely compensating score.

- A trusted global-identifier conflict vetoes `High` immediately and sends the pair to `Conflict Review`, even if weaker fields agree.
- An exact trusted global identifier may itself qualify the pair for `High`, but the record still carries warning metadata if names disagree.
- Exact names cannot become `High` on their own. The validated name-based path requires approved **independent evidence** (`birthday`, `phone`, or `email`) in addition to the exact full Chinese or English name.

7.3 Underlying reasoning model

Tiered Gated first classifies evidence by semantic role and only then decides the tier. In the implementation, each compared field yields an `Evidence` object with a level such as `exact`, `close`, `phonetic`, `weak`, `missing`, or `disagreeing`, while policy logic separately recognizes `trusted identifier`, `unverified identifier`, and `independent evidence` roles.

This produces a deterministic reasoning model with the following principles:

- Evidence types have different meanings and are **not** freely interchangeable.
- The model uses an **ordered decision tree**, not one compensating weighted score that sums every signal together.
- Repeated representations of the same name are not independent corroboration. Exact Chinese name and exact English name for the same person description still count as name evidence, not two separate safety approvals.
- Exact `birthday`, `phone`, or `email` is treated as independent corroboration for an exact full name because it comes from a different evidence family.
- `Missing` evidence is neutral: it withholds support, but it is distinct from an explicit disagreement.
- Safety-critical contradictions cannot be numerically cancelled by weaker agreements elsewhere.

For audit display, the Tiered Evidence Score is the **average of the available per-field evidence scores**. That average is descriptive only; it does not override the ordered policy rules.

7.4 Ordered decision priority

Order matters. The approved high-level priority is:

1. **Trusted global-ID conflict → `Conflict Review`**
2. **Exact trusted global ID → `High`**, possibly with a `name_conflict_warning`
3. **Exact full Chinese or English name plus exact birthday/phone/email → `High`**
4. **Useful name support, exact secondary evidence, or exact unverified identifier → `Review`**
5. **Otherwise → `Low`**

A pair that reaches `High` from step 2 or 3 still must pass the six downstream safety gates before a reversible `Proposed` recommendation is created.

7.5 Tiered Evidence Score versus Tiered Tier

The Tiered Evidence Score is **explanatory only**.

- It is **not a probability**.

- It does **not** control the final tier.
- The final tier comes from the ordered deterministic rules above.

Synthetic examples:

Synthetic pair pattern	Explanatory Tiered Evidence Score	Final tier	Why the tier wins over the score
Trusted HKID conflict + exact Chinese name + exact birthday + exact phone	0.90	`Conflict Review`	The trusted-ID contradiction is a hard gate and cannot be cancelled by other agreements
Exact full English name only; birthday/phone/ email all missing	0.84	`Review`	Exact name without approved independent corroboration is still review-only
Exact trusted HKID + disagreeing English surname + phone missing	0.58	`High`	The approved trusted-global-ID route takes priority; the disagreement is recorded as `name_conflict_warning` rather than downgrading the tier

7.6 Principle examples

Scenario	Evidence summary	Result
Exact full name plus exact birthday	Exact Chinese or English full name + exact birthday + no trusted-ID conflict	`High`
Exact full name only	Exact full name, but birthday/ phone/email provide no exact corroboration	`Review`
Trusted HKID conflict plus otherwise matching evidence	Disagreeing valid full HKIDs, even when name and birthday also match	`Conflict Review`
Exact unverified/local identifier	Exact local or out-of-scope identifier without trusted global scope	`Review` only
Missing birthday versus different birthday	Missing birthday is neutral and simply withholds support; a different birthday is explicit disagreement and weakens the case	Distinct evidence meanings; they are not treated the same

7.7 Principle-level comparison with Splink

Principle	Tiered Gated	Splink
Decision basis	Deterministic governed rules	Probabilistic learned model
What the model interprets	Approved evidence combinations and veto conditions	Likelihood of observed comparison patterns among same vs. different people
How contradictions are handled	Safety gates and ordered vetoes	Statistical match-weight combination across comparison patterns
Stability over time	Same inputs keep the same rule outcome until policy changes	Scores can change after retraining, recalibration, or comparison-definition changes
Approved operating role	Approved deterministic recommendation method	Approved only to rank an optional human-review queue at or above the maximum-F1 Review cutoff

The approved boundary remains unchanged: Tiered Gated is the only approved deterministic recommendation path, Splink feeds only the optional ranked human-review queue, and Hybrid, Tiered Recoverable, and the legacy baseline remain set aside as POC comparison or audit methods.

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8. Tiered Gated Decision Tree

The approved deterministic decision tree uses the **gated** conflict mode.

```
flowchart TD
    A[Pair enters Tiered Gated evaluation] --> B{Any trusted global ID<br/>disagrees?}
    B -- Yes --> CONFLICT[Conflict Review<br/>reason: trusted_global_id_conflict<br/>+
identifier_conflict_gate]
    B -- No --> C{Any trusted global ID<br/>matches exactly?}
    C -- Yes --> HIGHID[High route<br/>reason: trusted_global_id_exact<br/>possible
name_conflict_warning]
    HIGHID --> F
    C -- No --> D{Exact full Chinese<br/>or English name?}
    D -- Yes --> E{Any exact independent evidence:<br/>birthday / phone / email?}
    E -- Yes --> F[→ Safety Gates]
    E -- No --> REVIEW1[Review<br/>reason: human_review_required<br/>+
insufficient_independent_evidence]
    D -- No --> G{Useful name support,<br/>exact secondary evidence,<br/>or exact unverified ID?}
    G -- Yes --> REVIEW2[Review<br/>reason: human_review_required]
    G -- No --> LOW[Low<br/>reason: insufficient_evidence]
    F --> GATE1{Source-coverage gate}
    GATE1 -- Fail --> EX1[Exception]
    GATE1 -- Pass --> GATE2{Stale-data gate}
    GATE2 -- Fail --> EX2[Exception / Stale]
    GATE2 -- Pass --> GATE3{One-to-many gate}
    GATE3 -- Fail --> EX3[Exception]
    GATE3 -- Pass --> GATE4{Transitive-consistency gate}
    GATE4 -- Fail --> EX4[Exception]
    GATE4 -- Pass --> GATE5{Cluster-safety gate}
```

```
    GATE5 -- Fail --> EX5[Exception]
    GATE5 -- Pass --> HIGH[Proposed High Recommendation]
```

Key rules in the gated mode:

- **Any** non-empty trusted-identifier disagreement blocks the pair at `Conflict Review`. There is no recovery path in the approved operating mode.
- **Any** exact trusted global identifier match is its own High route. If name fields disagree, the pair remains High but carries `name_conflict_warning` for audit visibility.
- **No** comparison of name fields alone (even exact full names) is sufficient for High. The validated name-based High path always requires exact `birthday`, `phone`, or `email` corroboration.

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9. Reason Codes

Reason codes are stored on every recommendation and evaluation record. They explain the primary evidence that triggered the decision.

Reason code	Meaning
`trusted_global_id_exact:hkid`	Both sides have matching valid HKID with global scope
`trusted_global_id_exact:hksr_num`	Both sides have matching HKSR number with global scope
`trusted_global_id_conflict:hkid`	Non-empty valid HKIDs disagree
`trusted_global_id_conflict:hksr_num`	Non-empty HKSR numbers disagree
`chinese_full_name_exact`	Normalized Chinese surname + given name match exactly
`english_full_name_exact`	Normalized English surname + given name match exactly
`independent_exact:birthday`	Normalized birthdays match exactly
`independent_exact:phone`	Normalized phone numbers match exactly
`independent_exact:email`	Normalized email addresses match exactly
`unverified_identifier_exact:hkid`	HKIDs match but are not in global scope (review only; never automatic High on this path)
`unverified_identifier_exact:hksr_num`	HKSR numbers match but not in global scope (review only; never automatic High on this path)
`exact_name_plus_independent_evidence`	Exact full name plus exact birthday, phone, or email satisfied the validated name-based High route
`human_review_required`	Evidence is useful but insufficient for automatic High; route to Review
`insufficient_independent_evidence`	Exact name support exists, but approved independent corroboration is missing

Reason code	Meaning
`identifier_conflict_gate`	Trusted global identifier conflict forced `Conflict Review`
`name_conflict_warning`	Exact trusted global identifier produced High despite disagreeing name fields; retain warning for audit
`insufficient_evidence`	Evidence did not reach Review usefulness; route to Low
`source_coverage_exception`	Source-pair group not represented in validated cohort
`stale_data_exception`	One or more records modified after frozen snapshot
`one_to_many_source_conflict`	Record appears in multiple conflicting High edges
`transitive_inconsistency`	High edges in a component contradict each other
`cluster_safety_exception`	Component contains confirmed `Different` labels or conflicts

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10. Safety Gates

Safety gates are evaluated **after** the deterministic High rule is satisfied and **before** a `Proposed` recommendation is created. Every gate fails closed: a gate failure produces an `Exception` record rather than a recommendation.

10.1 Source-coverage gate

The pair's source combination (left source × right source) must appear in the approved High-validation cohort. Source-pair groups not represented in the 100-pair targeted validation sample remain exception-only until a separate validation is completed.

At the time of the current canary preview:

```
Governed source-pair groups:         All validated except three sparse groups
Sparse unvalidated groups:           3 (17 historical High candidates total)
Those 3 groups in current preview:   0 High candidates → 0 source-coverage exceptions
```

10.2 Stale-data gate

After the canary snapshot is captured, any CCD Master record that is subsequently modified (any field change, including administrative updates) is considered **stale**. Stale records produce an `Exception` rather than a `Proposed` recommendation.

The stale check compares the `source_modified` timestamp captured in the evaluation snapshot against the live `modified` value in CCD Master:

```
def _stale_record_ids(record_by_id, record_ids):
    # Re-fetch live modified timestamps from CCD Master
    # Compare against snapshot-time source_modified
    return {id for id in record_ids
            if current_modified[id] != snapshot_modified[id]}
```

10.3 One-to-many gate

A single CCD Master record must not appear in multiple High-tier edges pointing to different counterparts. If record A is High with both B and C, the system cannot safely recommend all three are the same person without additional evidence. The entire affected component becomes an `Exception` with reason `one_to_many_source_conflict`.

The current exception count is **433 exception edges across 191 connected-component cases** — all with reason `one_to_many_source_conflict`.

10.4 Transitive-consistency gate

Within a connected component of High edges, every implied identity relationship must be consistent. If A–B is High and B–C is High, then A–C must also be consistent. Internal contradictions fail the whole component.

10.5 Cluster-safety gate

A component that contains any confirmed `Different` human label for any pair within it fails this gate. Human-confirmed differences prevent automatic recommendation of any edge in the affected cluster.

10.6 Data-quality gate

Phone, email, and birthday values are normalized before becoming evidence. A value that fails normalization (e.g., a sequential placeholder phone number) produces a `MISSING` evidence level rather than a usable `EXACT` comparison. This gate is applied inline during normalization rather than as a separate gate step.

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11. Blocking Routes

Candidate pairs are generated through **blocking**: records sharing a common blocking key are candidates for comparison. Without blocking, comparing all 251,520 × 251,520 records would produce billions of comparisons.

11.1 Route priority order

```
BLOCK_ROUTE_PRIORITY = {
    "global_id":         0,   # highest priority
    "phone":             1,
    "email":             1,
    "unverified_id":     2,
    "dob_surname":       3,
    "chi_full":          4,
    "chi_pinyin_full":   5,
    "chi_given_sorted":  5,
    "eng_name":          6,
    "chi_name_prefix":   7,   # lowest priority (broadest)
}
```

Routes with lower numbers are added to the candidate pool first. The candidate pool is capped at `max_candidate_pairs = 500,000` per policy document.

11.2 Route descriptions

Route	Blocking key pattern	Purpose
`global_id`	`global_id:{attribute}:{value}`	Exact match on trusted, globally-scoped identifier
`phone`	`phone:{normalized_phone}`	Exact 8-digit phone after normalization
`email`	`email:{normalized_email}`	Exact email after normalization
`unverified_id`	`unverified_id:{attribute}:{value}`	Partial/local/unverified ID; for audit only
`dob_surname`	`dob_surname:{date}:{surname}`	Birthday combined with Chinese or English surname
`chi_full`	`chi_full:{surname}{given}`	Full concatenated Chinese name (exact)
`chi_pinyin_full`	`chi_pinyin_full:{surname}:{given_pinyin}`	Chinese surname + Pinyin of given name (covers homophones)
`chi_given_sorted`	`chi_given_sorted:{surname}:{sorted_chars}`	Chinese surname + sorted given-name characters (covers transpositions)
`eng_name`	`eng_name:{surname}:{given[:2]}`	English surname + first two characters of given name
`chi_name_prefix`	`chi_name_prefix:{surname}:{given[:1]}`	Chinese surname + first character of given name (broadest)

11.3 Oversized block skipping

Any blocking key that would produce a block larger than `max_block_size = 10,000` records is skipped. Skipped blocks are recorded by route and an anonymized hash of the key. The Splink Review queue requires **zero skipped blocks** and fails closed if any appear.

11.4 Broad-route pair selection

For `chi_name_prefix` and `eng_name` routes, the blocking step does not generate all combinations within a block. Instead, each record nominates its **closest-name counterpart per other source**, and nominations are ranked from the sparsest endpoints first. This prevents large integrated sources from consuming the entire candidate budget before sparse sources receive any candidates.

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12. Splink: Fellegi–Sunter Principles

Splink implements the **Fellegi–Sunter** probabilistic record linkage framework. The framework assigns a match probability to each candidate pair by comparing the likelihood that the observed field agreements and disagreements would occur if the two records represent the same person (match hypothesis M) vs. different people (non-match hypothesis U).

12.1 Match probability formula

For a pair with comparison observations $\gamma_1, \gamma_2, \ldots, \gamma_n$ across n comparison columns:

$$P(\text{Match} \mid \gamma_1 \ldots \gamma_n) = \frac{\lambda \times \prod m(\gamma_i) / u(\gamma_i)}{\lambda \times \prod m(\gamma_i)/u(\gamma_i) + (1-\lambda) \times 1}$$

Where:

- λ is the prior probability that any given candidate pair is a match (the random-match prior);
- $m(\gamma_i)$ is the probability that comparison γ_i is observed given the pair is a match;
- $u(\gamma_i)$ is the probability that comparison γ_i is observed given the pair is not a match.

12.2 Prior (λ)

The adapter uses Splink's conservative default random-match prior of **`0.0001`** (1 in 10,000). It does not estimate the prior from exact-name blocks, because common CCD names can create an inflated prior that falsely elevates probabilities for pairs that share a common name.

12.3 Comparison columns

The Splink model uses the following comparison columns (derived from the same evidence attributes as the deterministic model):

Comparison	Level hierarchy
Chinese full name	exact match / phonetically similar / first character match / else
English full name	exact match / similar / else
Phone	exact match / missing / else
Email	exact match / missing / else
Birthday	exact match / missing / else
HKID	exact match / missing / else

Missing values are represented as SQL nulls (not empty strings). This ensures Splink does not learn null/null agreement as evidence of identity.

13. m and u Probabilities

m probability: the probability that two fields in a matching pair agree at a given comparison level.
u probability: the probability that two fields in a non-matching pair agree at a given comparison level by chance.

13.1 Expectation–Maximization (EM) estimation

Splink estimates m and u probabilities iteratively using EM. Starting from initial estimates:

1. **E-step:** Use the current m/u estimates to compute a match probability for every training-cohort pair.
2. **M-step:** Use those probabilities as weights to re-estimate the m/u values.
3. Repeat until convergence.

EM uses up to **250,000 randomly sampled pairs** within the training cohort to estimate u probabilities. The same limit applies to the pairs used in each EM iteration.

13.2 Interpretation

For a phone comparison:

Scenario	m(exact)	u(exact)	Bayes factor
Typical pattern	~0.90	~0.001	~900

A Bayes factor of 900 means: an exact phone match makes the pair 900 times more likely to be a genuine match than a chance agreement.

13.3 Model version and parameters are frozen

The current model is `pilot-splink-1.1`. Its m/u estimates, comparison definitions, and field configurations are frozen for the duration of the current approved Review queue operation. Changes to any of these require a new model version and recalibration.

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14. Term Frequency Adjustment

Common names (such as `Chan` or `Lee`) are weaker identity evidence than rare names. Term-frequency (TF) adjustment reduces the match-probability contribution of a common-name agreement relative to a rare-name agreement.

Splink estimates term frequencies from the training-cohort population. For Chinese surname □ (Chan) which appears in many records:

```
TF adjustment for exact □ match: lower Bayes factor
TF adjustment for exact □□□ match: higher Bayes factor
```

Term frequencies are estimated once during training on the 5,000-record cohort. They reflect the population of that cohort. If the cohort underrepresents certain names, their frequency estimates may be less reliable.

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15. Training Cohort: 5,000-Record Bounded Design

15.1 Why 5,000 records

The ERPNext long worker has a fixed memory budget. Experiments confirmed that a 20,000-record training cohort with a 250,000-pair random-u sample exceeded the worker memory limit before scoring completed. The 20,000-record model was killed without producing any accuracy result.

```
5,000-record training:  Completed. Average precision 0.6242, ROC AUC 0.8714.
20,000-record training: Killed by memory limit. No accuracy result.
```

The 5,000-record limit is therefore a **hard operational constraint**, not an arbitrary choice.

15.2 What the 5,000 records contain

The training cohort is a **deterministic background sample** from the governed record population, supplemented by every record that appears as an endpoint in the retained human-review pairs. The selection is reproducible from the same frozen snapshot.

```
MAX_SPLINK_TRAINING_RECORDS = 5_000
```

15.3 What the 5,000-record limit does and does not mean

Statement	Correct?
Splink trains on 5,000 records	Yes
Splink can only score pairs involving those 5,000 records	No
The 5,000-record model can score pairs from the full 251,520-record population	Yes
The 5,000 records must represent all source systems	Aspirational; sparse sources may be underrepresented

The 5,000 records are used to fit the model (estimate m, u, and term frequencies). The fitted model is then applied to pairs drawn from the **full** governed population.

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16. Calibration vs. Training

These are distinct steps performed on different data:

Step	Data used	Purpose
Training	5,000-record background cohort	Estimate Splink m/u probabilities via EM
Calibration	Locked labeled evaluation pairs (calibration split)	Select an operating Review threshold
Validation		

Step	Data used	Purpose
	Locked labeled evaluation pairs (held-out split)	Estimate performance at the selected threshold

The calibration labels are human-reviewed pairs with `Same`, `Different`, or `Unsure` outcomes. The calibration step selects the cutoff that maximizes F1 on the calibration split. The held-out split is never used during threshold selection.

The current cutoff `0.938995074` was selected on the calibration split and confirmed on the held-out split of the `pilot-1.6` 500-pair representative recalibration.

---

17. Held-Out Validation

17.1 Split design

The 500-pair labeled evaluation cohort is split into:

- **Calibration split** (~70%): threshold selection
- **Held-out split** (~30%): independent performance estimate

Pairs from earlier human reviews are excluded from both splits to prevent label leakage.

17.2 POC held-out results at the approved cutoff

Metric	Value
Calibration precision	44.92%
Calibration recall	88.33%
Held-out precision	44.62%
Held-out recall	76.32%
Held-out F1	Calculated at maximum-F1 cutoff
Confirmed Same pairs above cutoff (held-out)	43/70
Confirmed Same pairs below cutoff (held-out)	27/70

The 27 confirmed Same pairs below the cutoff are not treated as `Different`. Their scores fall below the current Review-priority band; they are lower-priority, not excluded.

17.3 No valid automatic High threshold

The POC did not produce a Splink probability threshold that satisfies the automatic High precision target (Wilson 95% lower bound ≥ 95%). Splink remains a **review prioritization tool only**.

---

18. Review Queue: 20,000-Pair Batch Scoring

18.1 Architecture overview

The Review queue uses a **separate batch-scoring path** that does not share the old direct-scoring limit.

```
Training phase:
  5,000 training records → fit Splink model once

Scoring phase:
  251,520 governed records provide pair endpoints
  821,592 governed candidate pairs
  − 3,961 Tiered High pairs excluded
  − 1,097 previously human-used pairs excluded
  = 816,534 eligible pairs
  → scored in batches of 20,000 using DuckDB
  → 11,177 pairs at or above cutoff stored in Review queue
```

18.2 Why 20,000 pairs per batch

```
REQUESTED_PAIR_BATCH_SIZE = 20_000
```

Each batch loads its pairs into temporary DuckDB tables, joins exactly on pair sequence, scores, and discards the temporary tables. This avoids loading all 816,534 pairs into memory simultaneously.

```
816,534 ÷ 20,000 ≈ 41 scoring batches
```

18.3 Not a Cartesian join

A naive implementation would join 20,000 left records against 20,000 right records, producing 400,000,000 comparisons. The batch scorer does not do this.

Each pair in a batch is assigned an **opaque sequence number** (see Section 19). The DuckDB join condition is:

```
LEFT.__pair_sequence = RIGHT.__pair_sequence
```

This produces **exactly one comparison per requested pair** — 20,000 comparisons for 20,000 pairs.

18.4 Integrity verification

After scoring each batch, the adapter verifies:

```
Number of output scores = number of requested pairs in batch
```

Any batch with missing or duplicate pair scores causes the entire queue run to fail rather than publishing a partial result.

19. Opaque Pair Sequences and Distinct 5,000 Limits / Chunks

19.1 Opaque pair sequences

Each requested pair is assigned a deterministic integer sequence number before batch loading. The sequence number is opaque — it does not encode the record IDs or any identity-relevant data. It serves only as a join key within a single batch.

```
Pair (A, B) → sequence 0
Pair (C, D) → sequence 1
Pair (E, F) → sequence 2
…
```

After batch scoring, sequence numbers are discarded. The output maps back to pair keys `(left_id, right_id)`.

19.2 Distinct 5,000 limits

There are three separate limits of 5,000 that must not be confused:

Limit	Value	Meaning	Replaced by batch mode?
Training cohort size	5,000 records	Maximum records used to fit the Splink model	No — still applies
Direct-scoring pair limit	5,000 pairs	Maximum pairs scored via `compare_two_records()` one at a time	Yes, for the batch queue path
Database insert chunk	5,000 rows	Number of queue rows written per database insert batch	No — this is an insert batch, not a queue cap

```
MAX_DIRECT_SCORING_PAIRS = 5_000   # still applies when batch_requested_pairs=False
REQUESTED_PAIR_BATCH_SIZE = 20_000 # scoring batch in the Review queue path
```

19.3 When direct scoring is still used

Direct scoring via `compare_two_records()` is still used when `batch_requested_pairs=False`. This applies to:

- bounded evaluation samples (≤ 500 pairs);
- repair runs for a small held-out validation set;
- any situation where the requested pairs were not generated through the governed batch pipeline.

The 5,000 direct-scoring limit protects against accidentally calling `compare_two_records()` hundreds of thousands of times through the old path.

20. Snapshot Reproducibility and Fail-Closed Safeguards

20.1 The frozen canary snapshot

The canary captures a frozen population at the time of its creation:

- `record_count`: exact number of governed CCD Master records in the snapshot
- `candidate_count`: exact number of governed candidate pairs
- `snapshot_at`: timestamp

A Splink Review queue created from that canary must reproduce the **entire** frozen snapshot, not only the 5,000-record training cohort.

20.2 Explicit fail-closed checks

The Review queue run includes the following explicit fail-closed checks:

```
# Check 1: Stale or deleted snapshot records
if stale_snapshot_records or len(records) != int(canary.record_count or 0):
    frappe.throw(
        "The frozen canary record population is no longer reproducible"
    )

# Check 2: Candidate generation must be complete (no truncation, no skipped blocks)
if blocked.truncated or blocked.skipped_blocks:
    frappe.throw(
        "Splink Review queue requires complete candidate generation"
    )

# Check 3: Regenerated candidate count must match frozen canary
if len(blocked.pairs) != int(canary.candidate_count or 0):
    frappe.throw(
        "The regenerated candidates differ from the frozen canary"
    )

# Check 4: Stale training cohort
if stale_training:
    frappe.throw(
        "The approved Splink training cohort changed; recalibration is required"
    )
```

These checks are **code-level** fail-closed gates. They abort the queue run if any reproducibility condition is not met. There is no UI equivalent: the system refuses to produce a partial or inconsistent queue.

20.3 Why full snapshot reproducibility is required

The Review cutoff `0.938995074` was validated against the **specific frozen population** used in the canary. Scoring a different population (even if only a few records changed) with the same cutoff produces probabilities that cannot be compared against the validated threshold.

A queue created long after its canary must prove all of the following before it will run:

1. Every record in the canary snapshot still exists in CCD Master;
2. No record has been modified since the snapshot;
3. The reproduced candidate-pair count exactly matches the frozen count;

4. The 5,000-record training cohort endpoints are all still unmodified;
5. No candidate-generation block was truncated or skipped;
6. The same policy snapshot, Splink adapter, and cutoff are active.

---

21. Pair Safety vs. Cluster Safety

21.1 Pair-level safety

A pair passes pair-level safety when:

- the deterministic High rule is satisfied;
- neither record is stale;
- the source-pair group is covered.

This is necessary but not sufficient for a recommendation.

21.2 Cluster-level safety

Pairs are part of connected components (clusters) of High edges. Before any edge in a cluster becomes a `Proposed` recommendation, the **full cluster** must pass:

- **One-to-many check:** no record appears in multiple edges pointing to different counterparts
- **Transitive-consistency check:** all implied relationships are mutually consistent
- **Cluster-safety check:** no confirmed `Different` human label exists for any pair in the cluster

A single failing edge blocks the **entire component** from receiving recommendations.

21.3 Current exception statistics

```
Total High edges before gates:     3,528 Proposed + 433 Exception
Exception edges:                   433
Exception components:              191
Exception reason:                  All 433 = one_to_many_source_conflict
```

No exceptions due to source-coverage, stale-data, transitive, or cluster-safety failures in the current preview. All exception edges are currently in the one-to-many category.

---

22. Review Workflow

22.1 Deterministic High recommendations

```
1. Canary run generates Proposed recommendations
2. Each recommendation stores: model version, evidence reasons, snapshot time,
   source scope, status, and immutable reversal history
3. Periodic random QC sample (~100 pairs) is drawn from Proposed
4. QC reviewers confirm Same or Different independently
5. Aggregate approval step activates the full Proposed batch
6. Active recommendations are available for operational use
7. Exception queue presents 191 connected-component cases for human resolution
```

22.2 Splink Review queue

```
1. Queue generates 11,177 candidates ranked by Splink probability (descending)
2. Candidates are an optional, capacity-based pool
3. Reviewer assignment must be limited by operational capacity
4. Ordinary reviewers see masked values; no record IDs or Splink scores
5. Sensitive Reviewers / System Managers see full-value links
6. Individual probabilities are System Manager-only
7. Same requires two independent confirmations from different reviewers
8. Unsure and reviewer disagreements require adjudication
9. Human-confirmed Same pairs are stored separately from model High predictions
10. All reviewer decisions are immutable and auditable
```

22.3 Role separation

Role	Sees
Ordinary reviewer	Masked field values; no record IDs, no model scores
Sensitive Reviewer	Full field values, record links
System Manager	All above plus individual Splink probabilities

Ordinary reviewers do not see model tier labels or reason codes during the review phase. This prevents anchoring on the model's assessment.

---

23. Approved vs. Not-Approved Summary Table

Capability	Approved?	Notes
Tiered Gated High recommendations	**Yes**	Reversible Proposed only; no automatic activation
Reversible recommendation records	**Yes**	`Proposed` → `Active` requires separate aggregate approval
Splink Review queue for human prioritization	**Yes**	Optional; capacity-based; no automatic decisions
Splink cutoff `0.938995074` for review priority	**Yes**	Snapshot-specific; must recalibrate on any material change
Record merging	**No**	Out of scope until canary reviewed
Automatic `Is Matched?` setting	**No**	Out of scope until canary reviewed
Probabilistic automatic High threshold	**No**	POC did not produce a validated threshold
Exact trusted global-ID High path	**Yes**	Approved deterministic High route; may carry

Capability	Approved?	Notes
		`name_conflict_warning` when name fields disagree
Tiered Recoverable conflict path to High	**No**	POC method; temporarily set aside
Hybrid deterministic+probabilistic High	**No**	POC method; temporarily set aside
Legacy baseline formula as identity decision	**No**	Remains as control/current path only
20,000-record Splink training	**No**	Exceeded worker memory limit; no accuracy result

---

24. Snapshot-Specific POC Examples (Sanitized)

The following examples use synthetic data that does not correspond to any real CCD Master record or client. They illustrate how the approved decision tree behaves.

24.1 Example A — High via Chinese name + phone

```
Left record:   Chi name = □□□ | Phone = 91234567 | Source = S-01
Right record:  Chi name = □□□ | Phone = 91234567 | Source = S-04

Normalization:
  Chi name left/right:  □□□ / □□□  → exact
  Phone left/right:     91234567 / 91234567  → exact

Evidence:
  chinese_full_name_exact   = True
  independent_exact:phone   = True
  trusted_global_id_conflict = None

Safety gates: source-coverage (S-01×S-04 validated), stale (no), one-to-many (no), transitive (N/A), cluster-safety (no prior Different)

Result: Proposed High
Reason codes: chinese_full_name_exact, independent_exact:phone, exact_name_plus_independent_evidence
```

24.2 Example B — Conflict Review

```
Left record:  HKID = A123456(7) | Source = S-02 (scope = global)
Right record: HKID = B234567(8) | Source = S-03 (scope = global)

Normalization:
  Left HKID:  A1234567 → valid, global scope
  Right HKID: B2345678 → valid, global scope

Evidence:
  trusted_global_id_conflict:hkid = True
```

```
Result: Conflict Review (regardless of name or other evidence)
Reason codes: trusted_global_id_conflict:hkid, identifier_conflict_gate
```

24.3 Example C — Review (name only)

```
Left record:   Chi name = □□□ | Phone = 98765432 | Source = S-01
Right record:  Chi name = □□□ | Phone = missing  | Source = S-05

Evidence:
  chinese_full_name_exact   = True
  independent_exact:phone   = False (right side missing)
  independent_exact:birthday = False (both missing)
  independent_exact:email    = False (both missing)

Result: Review
Reason codes: chinese_full_name_exact, human_review_required, insufficient_independent_evidence
```

24.4 Example D — Stale exception

```
Canary snapshot captured: 2026-08-01 09:00:00
Left record modified:     2026-08-02 14:23:11  (after snapshot)

Stale gate: modified timestamp mismatch detected

Result: Exception
Reason code: stale_data_exception
```

24.5 Example E — Below Splink cutoff (lower priority, not Different)

```
Candidate pair: eligible for Splink scoring (not Tiered High, not human-used)
Splink score:   0.72

Cutoff: 0.938995074
0.72 < 0.938995074

Result: NOT stored in Review queue
Interpretation: Lower human-review priority
This pair is NOT labeled Different; it remains an unreviewed candidate
```

---

25. Audit and Research Methods (Temporarily Set Aside)

The following methods were evaluated during the POC as comparison baselines. They are **not** part of the approved operating path. They are retained in the codebase for audit, research, and potential future investigation.

25.1 Tiered Recoverable

The Recoverable variant allowed independent name and exact secondary evidence to recover a trusted-identifier conflict pair from `Conflict Review` to ordinary `Review` (never directly to `High`). This was evaluated to determine whether strong secondary evidence should override a possible data-entry error in an identifier.

POC conclusion: The approach is useful for routing some conflicts to ordinary Review rather than the dedicated Conflict Review queue. However, management approved the Gated approach for the current operating phase. Recoverable is not part of the approved High path.

25.2 Hybrid (Splink + Deterministic Gates)

The Hybrid approach applied deterministic identifier safety gates around a calibrated Splink score. It was intended to use a probabilistic threshold for High decisions after the deterministic gates passed.

POC conclusion: The POC did not produce a validated Splink probabilistic High threshold. Without a validated threshold, Hybrid cannot add automatic High decisions. It remains shadow/review-prioritization only.

25.3 Legacy Baseline Formula

The existing `fuzzymachingscript` weighted formula is retained as the control path. It evaluates each registration's current fuzzy logic in its own direction and reports the maximum directional score.

POC conclusion: The current flagged pairs from the baseline had low precision. The score is not a calibrated identity probability. No new automatic decisions are authorized from the baseline formula. It is retained as the current/control path for comparison purposes.

---

26. Recalibration Triggers

The snapshot-specific cutoff `0.938995074` is tied to a specific set of conditions. **Any** of the following changes requires recalibration before the cutoff can be used for a new Review queue:

Change trigger	Reason
Splink adapter version changes	m/u estimates and comparison definitions may differ
Training cohort changes	Term frequencies and m/u estimates depend on the training population
Comparison field definitions change	Changing what constitutes "exact", "close", etc. changes all probabilities
Normalization rules change	Different normalization produces different comparison outcomes
Policy document changes	Source profiles, trusted identifiers, or aliases affect which fields are compared
Labeled calibration data significantly expands	Better labels may move the optimal F1 threshold
CCD Master population materially changes	New sources, data migration, or large import changes the candidate population
Canary snapshot changes	The frozen population for which the cutoff was validated has changed

Process for recalibration:

1. Freeze a new training cohort with the same 5,000-record limit

2. Fit a new Splink model (assign a new adapter version, e.g., `pilot-splink-1.2`)
3. Score the locked labeled calibration pairs with the new model
4. Select the new maximum-F1 cutoff on the calibration split
5. Validate on the held-out split
6. Record the new cutoff, adapter version, and cohort snapshot
7. Submit for management approval before activating a new Review queue

The old cutoff `0.938995074` must not be reused with a new model version, even if the numeric value happens to seem similar.

---

27. Wilson 95% Confidence Interval

27.1 Origin

The Wilson score interval was introduced by **Edwin Bidwell Wilson** (1879–1964) in a 1927 paper titled *Probable Inference, the Law of Succession, and Statistical Inference*. It is a standard method for computing confidence intervals for proportions, particularly when the sample proportion is near 0% or 100%.

27.2 POC result

The targeted 100-pair High-only validation cohort produced:

```
Correct (Same) predictions:   100
Total predictions reviewed:   100
Observed precision:           100%

Wilson 95% lower bound:       96.30%
Wilson 95% upper bound:       100%
Policy precision target:      95%

96.30% > 95%  → acceptance criterion passed
```

27.3 Interpretation

The Wilson interval expresses sampling uncertainty. Even with 100/100 correct results, a sample of 100 does not prove that all future predictions will be correct. The lower bound of 96.30% is the conservative estimate: if the same sampling procedure were repeated many times, approximately 95% of the resulting intervals would contain the true precision.

27.4 Why Wilson is preferred over simple ±

A simple normal-approximation interval (`p ± 1.96 × √(p(1-p)/n)`) behaves poorly when `p = 1.0` (zero observed failures). It would produce an upper bound of exactly 100% and a lower bound that is mathematically undefined or zero. Wilson's method handles this case correctly.

27.5 Sample size effect

Correct / sample	Approx. Wilson 95% lower bound
10 / 10	72.25%
30 / 30	88.65%

Correct / sample	Approx. Wilson 95% lower bound
50 / 50	92.86%
73 / 73	95.00%
100 / 100	96.30%

The POC used 100 pairs specifically to ensure that a perfect result still exceeded the 95% policy target at the lower confidence bound.

27.6 What the interval measures

The Wilson interval estimates **precision** (of the sampled High predictions), not:

- recall (proportion of all true duplicates that were found);
- precision of Review predictions;
- precision of Splink probabilities;
- precision of the production baseline formula.

---

28. Glossary

Term	Definition
Tiered Gated	The approved deterministic recommendation method; trusted-ID conflicts are gated to Conflict Review with no recovery path
Tiered Recoverable	POC comparison method; conflict recovery to Review via secondary evidence; temporarily set aside
High	A pair that passes the approved deterministic rule and all six safety gates; emitted as Proposed only
Review	A pair with useful but insufficient evidence for automatic High; requires human evaluation
Conflict Review	A pair with disagreeing trusted global identifiers; requires human resolution
Low / Insufficient	A pair with little or no useful shared evidence
Proposed	Recommendation status after all gates pass; not yet activated
Active	Recommendation status after aggregate approval
Exception	A High-eligible pair blocked by a safety gate
Stale	A record or recommendation edge whose underlying CCD Master data changed after the snapshot
Splink	

Term	Definition
	Open-source Fellegi–Sunter probabilistic record linkage library; used here for review prioritization only
pilot-splink-1.1	The frozen Splink adapter version used for the approved Review queue
Review cutoff	The Splink match probability threshold for entering the ranked human-review queue; currently `0.938995074`
m probability	The probability that a comparison outcome is observed given the pair is a genuine match
u probability	The probability that a comparison outcome is observed by chance given the pair is not a match
Bayes factor	`m(γ) / u(γ)`; how much a comparison outcome updates the prior match odds
Term frequency	How common a specific value (e.g., a surname) is in the training population; used to adjust match weight
Training cohort	The bounded 5,000-record population used to fit the Splink model; distinct from scoring records
Calibration labels	Human-reviewed pairs used to select the Review threshold
Held-out split	Labeled pairs reserved for independent performance estimation; not used during threshold selection
Wilson 95% CI	A confidence interval for a proportion that handles near-0% and near-100% results correctly; named after Edwin B. Wilson
EM (Expectation–Maximization)	The algorithm Splink uses to iteratively estimate m/u probabilities
Blocking	The process of reducing comparison space by only comparing pairs that share a common blocking key
Opaque pair sequence	An integer join key assigned to each pair within a scoring batch; does not encode identity
Fail-closed	A safety property where any unexpected condition aborts the operation rather than producing a partial result
Snapshot reproducibility	The property that a canary run created later can regenerate exactly the same record population and candidate pairs

Term	Definition
Source-pair group	A combination of two governed source systems (e.g., S-01 × S-04); validated separately
SourceProfile	A policy object describing one source system's field mapping, identifier scope, and disabled attributes
Reliability Status	A CCD Master field not used as evidence due to inconsistent population; stale-data gate is used instead

---

29. Architecture Diagrams

29.1 Approved operating path overview

```
flowchart TD
    A[Latest governed CCD candidate pair] --> B{Tiered Gated rule passes?}
    B -- No --> C[Review / Conflict / Low]
    B -- Yes --> D{Six safety gates pass?}
    D -- Any fail --> E[Exception record]
    D -- All pass --> F[Proposed High recommendation]
    F --> G{Aggregate approval}
    G -- Approved --> H[Active recommendation]
    G -- Not yet --> F
    H --> I[Periodic QC sample]
    I --> J[Human review: Same / Different]

    K[Governed candidate pair eligible for Splink] --> L[Splink batch scoring]
    L --> M{Score ≥ 0.938995074?}
    M -- Yes --> N[Ranked Review queue: higher priority]
    M -- No --> O[Lower review priority — NOT Different]
```

29.2 Splink batch scoring architecture

```
flowchart LR
    A[5,000 training records] -->|Fit once| B[pilot-splink-1.1 model]
    C[251,520 governed records] --> D[821,592 candidate pairs]
    D -->|Exclude 3,961 Tiered High| E[817,631 pairs]
    E -->|Exclude 1,097 human-used| F[816,534 eligible pairs]
    F -->|Split into batches of 20,000| G[~41 scoring batches]
    B --> G
    G -->|Score exactly one per pair| H[816,534 scored pairs]
    H -->|Retain ≥ 0.938995074| I[11,177 Review queue candidates]
```

29.3 Fail-closed snapshot reproducibility checks

```
flowchart TD
    A[Review queue run triggered] --> B{Stale/deleted snapshot records?}
    B -- Yes --> FAIL1[FAIL: population not reproducible]
    B -- No --> C{Record count = frozen canary count?}
    C -- No --> FAIL2[FAIL: population not reproducible]
    C -- Yes --> D{Candidate generation complete?<br/>No truncation, no skipped blocks?}
```

```
    D -- No --> FAIL3[FAIL: complete candidates required]
    D -- Yes --> E{Regenerated candidate count<br/>= frozen canary candidate count?}
    E -- No --> FAIL4[FAIL: candidates differ from canary]
    E -- Yes --> F{Training cohort endpoints stale?}
    F -- Yes --> FAIL5[FAIL: recalibration required]
    F -- No --> G[Proceed to Splink training and scoring]
```

29.4 Tiered Gated evidence requirement summary

```
flowchart LR
    A[Pair] --> B{Trusted ID conflict?}
    B -- Yes --> CONFLICT[Conflict Review]
    B -- No --> C{Exact trusted global ID?}
    C -- Yes --> HIGHID[High route<br/>possible name_conflict_warning]
    HIGHID --> F
    C -- No --> D{Exact full name?<br/>Chinese OR English}
    D -- No --> NOHIGH[Review or Low]
    D -- Yes --> E{Exact independent evidence?<br/>Phone OR Birthday OR Email}
    E -- No --> REVIEW[Review]
    E -- Yes --> F{Source-coverage validated?}
    F -- No --> EX1[Exception]
    F -- Yes --> G{No stale records?}
    G -- No --> EX2[Exception/Stale]
    G -- Yes --> H{One-to-many / Transitive / Cluster safe?}
    H -- No --> EX3[Exception]
    H -- Yes --> HIGH[Proposed High]
```

This guide reflects the management-approved post-POC operating baseline as of 2026-08-19. All statements are grounded in the current repository code and POC documentation. No real client data or source-system identifiers are included.